

OLLSCOIL NA hÉIREANN, CORCAIGH
THE NATIONAL UNIVERSITY OF IRELAND, CORK

COLÁISTE NA hOLLSCOILE, CORCAIGH
UNIVERSITY COLLEGE, CORK

Examination Session and Year	Winter 2023								
Module Code	EC4215								
Module Title	Business Econometrics and Forecasting								
Paper Number	1								
External Examiner	Prof. Mac an Bhaird								
The Head of the Department	Prof. Doran								
Internal Examiners	Dr. Turner								
Instructions to Candidates	<table><tr><td>Sections</td><td>No. of Questions to be attempted</td><td>% of total marks available</td></tr><tr><td>A</td><td>Long Questions 2 out of 3</td><td>100%</td></tr></table>			Sections	No. of Questions to be attempted	% of total marks available	A	Long Questions 2 out of 3	100%
	Sections	No. of Questions to be attempted	% of total marks available						
	A	Long Questions 2 out of 3	100%						
	All questions within each section carry equal weight								
	Attempt the correct number of questions in each section: Only that number of questions will be graded. If you attempt more than the correct number, please indicate what questions you wish to have graded. If you fail to do this, the appropriate number of <u>lowest-numbered questions</u> of those you answered will be graded.								
Duration of Paper	1.5 hours								
Special Requirements	Non-programmable calculators may be used								

**PLEASE DO NOT TURN THIS PAGE UNTIL INSTRUCTED TO DO SO
THEN ENSURE THAT YOU HAVE THE CORRECT EXAM PAPER**

Section A: Long Questions (Answer 2 out of 3)

- A1. (a)** Suppose that a researcher is examining the daily demand for ice cream cones at a seaside shop in Cork over the course of a summer and her data produces the following econometric regression results (standard errors in brackets):

$$C_t = 12.4 + 21.4T_t - 8.5R_t + 15.8W_t \quad (\text{Eqn. 1.1})$$

(6.4) (2.3) (5.1)

$$\bar{R}^2 = 0.68 \quad n = 92$$

Where:

C_t = the number of ice cream cones sold in the shop on day t ;

T_t = the maximum recorded temperature in Cork on day t (in degrees Celsius);

R_t = the total rainfall in Cork on day t (in millimetres);

W_t = a dummy variable set to 1 if the day falls at a weekend and 0 otherwise.

- (i)** Carefully interpret the above findings. **(35 per cent)**
- (ii)** Does the above equation fit your expectations? Why or why not? **(15 per cent)**
- (iii)** If you could add one more variable into the above regression, what would it be and why? **(10 per cent)**
- (b)** Suppose that the Minister for Agriculture asks you to use regression analysis to examine the factors that affect dairy output in Ireland.
- (i)** Explain in detail how you would proceed to carry out the requested analysis. **(25 per cent)**
- (ii)** How would you obtain your data sample for this specific exercise and how would you ensure that it is a suitable sample? Explain your answer. **(15 per cent)**

- A2. (a)** Suppose a researcher has run a regression to examine the factors affecting the number of domestic solar panel installations in Ireland over time and gets the following results (standard errors in brackets):

$$SP_t = 45.1 + 22.7 PE_t + 0.47 EV_t + 57.8 DI_t \quad (Eqn. 2.1)$$

(4.5) (0.15) (23.4)

$N = 36$ Adjusted $R^2 = 0.715$

Where:

SP_t = the number of domestic solar panel installations in Ireland in month t

PE_t = the average unit rate for electricity in month t (in cent per kWh)

EV_t = the number of electric vehicles sold in Ireland in month t

DI_t = the average household disposable income in Ireland in month t (€000)

Suppose the researcher is debating whether to include a variable to assess the effectiveness of the government grant for solar panel installations. If the researcher adds in this variable, she gets the following results:

$$SP_t = 36.8 + 22.6 PE_t + 0.47 EV_t + 57.6 DI_t + 2.2 GG_t \quad (Eqn. 2.2)$$

(4.9) (0.18) (27.1) (1.6)

$N = 36$ Adjusted $R^2 = 0.682$

Where:

GG_t = the level of the government grant in month t (€000)

All of the other variables are as before.

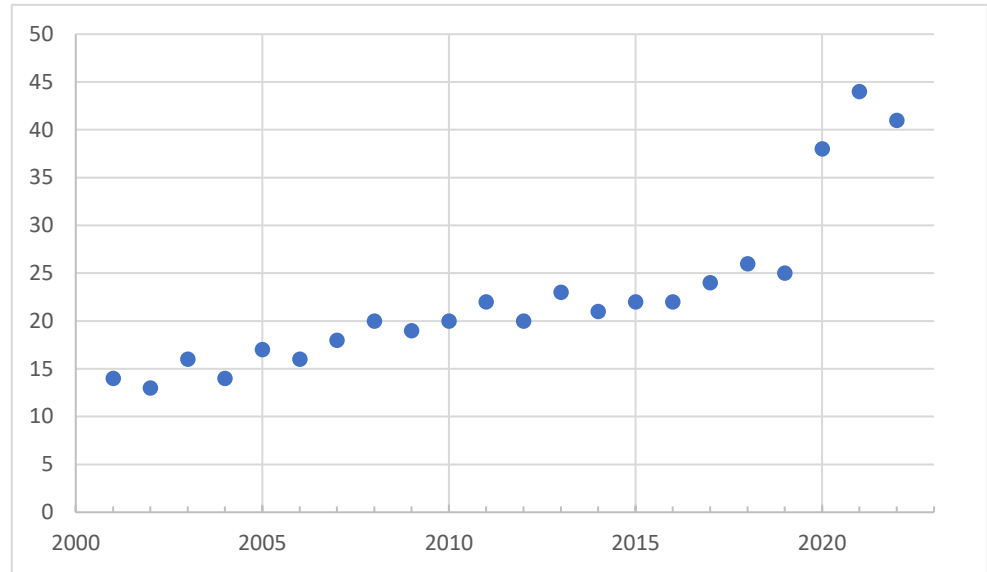
Should the researcher include GG_t in her regression (i.e. go with Eqn. 2.2 instead of Eqn. 2.1)? Explain your answer.

(70 per cent)

Question A2 continued on next page

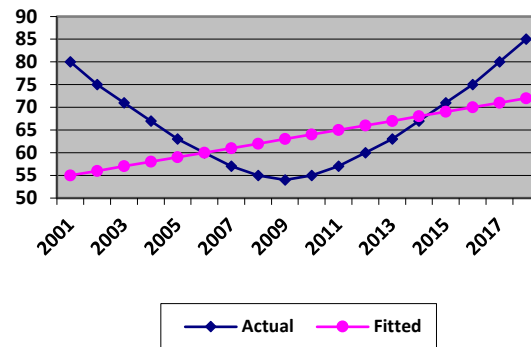
Question A2 continued

- (b) Suppose that you are given an annual time series dataset for a dependent variable from 2001 to 2022 inclusive, which looks like the one below.

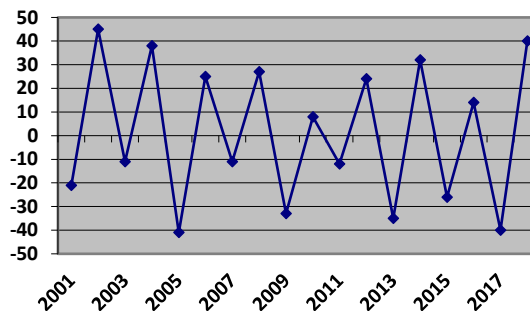


- (i) What potential specification issue is exemplified by this graph and what are the consequences of this issue?
(15 per cent)
- (ii) If you were told that the dependent variable is the value of online sales in Ireland, how would you deal with this specification issue in this case?
(15 per cent)

- A3. (a)** Suppose that a researcher runs a regression on a time-series dataset from 2001 to 2018 and gets the following actual and fitted values:



- (i) Is this regression likely to suffer from any econometric problem, and if so, which one?
(15 per cent)
- (ii) What is the likely cause of this econometric problem in this case? Explain your reasoning.
(15 per cent)
- (b) Below is a graph of residuals from a time-series regression.



- (i) What does this graph of residuals suggest? Explain your answer.
(15 per cent)
- (ii) Explain, with the aid of an example, why this might occur.
(15 per cent)

Question A3 continued on next page

Question A3 continued

- (c) For each of the following scenarios, state whether autocorrelation is present, based on the Durbin Watson test with a 5% level of significance. Briefly explain your answer.

- (i) $d = 1.56, n = 25, k = 3$
- (ii) $d = 2.89, n = 30, k = 5$
- (iii) $d = 2.35, n = 20, k = 4$
- (iv) $d = 1.55, n = 40, k = 2$

(20 per cent)

- (d) Suppose a researcher wants to examine trends in smoking among teenagers over time, and collects quarterly data on a number of variables, proposing the following model:

$$SR_t = \beta_0 + \beta_1 INC_t + \beta_2 AS_t + \beta_3 SR_{t-1} + \beta_4 GDP_t + \varepsilon_t$$

Where:

SR_t = the smoking rate (as a percentage) among teenagers in quarter t

INC_t = the average weekly income of teenagers (in €) in quarter t

AS_t = the amount spent on cigarette advertising (in €m) in quarter t

GDP_t = the value of Gross Domestic Product (in €m) in quarter t

ε_t = a stochastic error term

Given that this is a time series regression, the researcher believes it may suffer from autocorrelation and wants to test this formally. How would you recommend the researcher goes about this in this case? Justify your answer.

(20 per cent)

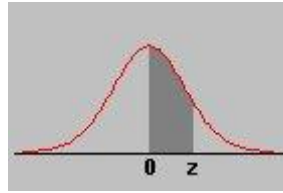
Appendix

A1: Selected Formulae

Slope coefficient: $\hat{\beta}_1 = \frac{\sum(X - \bar{X})(Y - \bar{Y})}{\sum(X - \bar{X})^2}$	Intercept coefficient: $\hat{\beta}_0 = \bar{Y} - \hat{\beta}_1 \bar{X}$
Standard Error of slope coefficient: $se(\hat{\beta}_1) = \frac{\hat{\sigma}}{\sqrt{\sum x_i^2}}$	Standard Error of intercept coefficient: $se(\hat{\beta}_0) = \sqrt{\frac{\sum X_i^2}{n \sum x_i^2}} \hat{\sigma}$
Standard Error of estimated regression: $\hat{\sigma} = \sqrt{\frac{\sum e_i^2}{n - k - 1}}$	Total, Explained and Residual Sum of Squares: $\sum_{i=1}^n (Y_i - \bar{Y})^2 = \sum_{i=1}^n (\hat{Y}_i - \bar{Y})^2 + \sum_{i=1}^n e_i^2$
<u>Confidence Interval:</u> Confidence Interval = $\hat{\beta}_k \pm t_{crit} * se(\hat{\beta}_k)$	<u>t-test:</u> $t = \frac{\hat{\beta}_k - \beta_{H0}}{se(\hat{\beta}_k)} \sim t(n - k - 1) df$
<u>F test (overall significance):</u> $F = \frac{ESS/k}{RSS/(n - k - 1)} \sim F(k, n - k - 1)$	<u>F test (overall significance):</u> $F = \frac{R^2/(k)}{(1 - R^2)/(n - k - 1)} \sim F(k, n - k - 1)$
<u>F test (Linear Restrictions):</u> $F = \frac{(RSS_R - RSS_{UR})/m}{RSS_{UR}/(n - k - 1)} \sim F(m, n - k - 1)$	<u>F test (Linear Restrictions):</u> $F = \frac{(R_{UR}^2 - R_R^2)/m}{(1 - R_{UR}^2)/(n - k - 1)} \sim F(m, n - k - 1)$

A2: Areas Under Normal Curve

Area between 0 and z

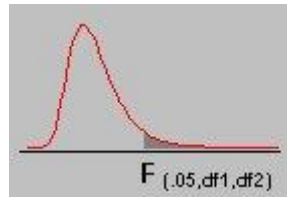


Z	0.00	0.01	0.02	0.03	0.04	0.05	0.06	0.07	0.08	0.09
0.0	0.0000	0.0040	0.0080	0.0120	0.0160	0.0199	0.0239	0.0279	0.0319	0.0359
0.1	0.0398	0.0438	0.0478	0.0517	0.0557	0.0596	0.0636	0.0675	0.0714	0.0753
0.2	0.0793	0.0832	0.0871	0.0910	0.0948	0.0987	0.1026	0.1064	0.1103	0.1141
0.3	0.1179	0.1217	0.1255	0.1293	0.1331	0.1368	0.1406	0.1443	0.1480	0.1517
0.4	0.1554	0.1591	0.1628	0.1664	0.1700	0.1736	0.1772	0.1808	0.1844	0.1879
0.5	0.1915	0.1950	0.1985	0.2019	0.2054	0.2088	0.2123	0.2157	0.2190	0.2224
0.6	0.2257	0.2291	0.2324	0.2357	0.2389	0.2422	0.2454	0.2486	0.2517	0.2549
0.7	0.2580	0.2611	0.2642	0.2673	0.2704	0.2734	0.2764	0.2794	0.2823	0.2852
0.8	0.2881	0.2910	0.2939	0.2967	0.2995	0.3023	0.3051	0.3078	0.3106	0.3133
0.9	0.3159	0.3186	0.3212	0.3238	0.3264	0.3289	0.3315	0.3340	0.3365	0.3389
1.0	0.3413	0.3438	0.3461	0.3485	0.3508	0.3531	0.3554	0.3577	0.3599	0.3621
1.1	0.3643	0.3665	0.3686	0.3708	0.3729	0.3749	0.3770	0.3790	0.3810	0.3830
1.2	0.3849	0.3869	0.3888	0.3907	0.3925	0.3944	0.3962	0.3980	0.3997	0.4015
1.3	0.4032	0.4049	0.4066	0.4082	0.4099	0.4115	0.4131	0.4147	0.4162	0.4177
1.4	0.4192	0.4207	0.4222	0.4236	0.4251	0.4265	0.4279	0.4292	0.4306	0.4319
1.5	0.4332	0.4345	0.4357	0.4370	0.4382	0.4394	0.4406	0.4418	0.4429	0.4441
1.6	0.4452	0.4463	0.4474	0.4484	0.4495	0.4505	0.4515	0.4525	0.4535	0.4545
1.7	0.4554	0.4564	0.4573	0.4582	0.4591	0.4599	0.4608	0.4616	0.4625	0.4633
1.8	0.4641	0.4649	0.4656	0.4664	0.4671	0.4678	0.4686	0.4693	0.4699	0.4706
1.9	0.4713	0.4719	0.4726	0.4732	0.4738	0.4744	0.4750	0.4756	0.4761	0.4767
2.0	0.4772	0.4778	0.4783	0.4788	0.4793	0.4798	0.4803	0.4808	0.4812	0.4817
2.1	0.4821	0.4826	0.4830	0.4834	0.4838	0.4842	0.4846	0.4850	0.4854	0.4857
2.2	0.4861	0.4864	0.4868	0.4871	0.4875	0.4878	0.4881	0.4884	0.4887	0.4890
2.3	0.4893	0.4896	0.4898	0.4901	0.4904	0.4906	0.4909	0.4911	0.4913	0.4916
2.4	0.4918	0.4920	0.4922	0.4925	0.4927	0.4929	0.4931	0.4932	0.4934	0.4936
2.5	0.4938	0.4940	0.4941	0.4943	0.4945	0.4946	0.4948	0.4949	0.4951	0.4952
2.6	0.4953	0.4955	0.4956	0.4957	0.4959	0.4960	0.4961	0.4962	0.4963	0.4964
2.7	0.4965	0.4966	0.4967	0.4968	0.4969	0.4970	0.4971	0.4972	0.4973	0.4974
2.8	0.4974	0.4975	0.4976	0.4977	0.4977	0.4978	0.4979	0.4979	0.4980	0.4981
2.9	0.4981	0.4982	0.4982	0.4983	0.4984	0.4984	0.4985	0.4985	0.4986	0.4986
3.0	0.4987	0.4987	0.4987	0.4988	0.4988	0.4989	0.4989	0.4989	0.4990	0.4990

A3: Student t- Distribution

d.f.	Level of Significance for One-Tailed Test						
	0.1	0.05	0.025	0.01	0.005	0.001	0.0005
	Level of Significance for Two-Tailed Test						
	0.2	0.1	0.05	0.02	0.01	0.002	0.001
1	3.078	6.314	12.706	31.821	63.656	318.289	636.578
2	1.886	2.920	4.303	6.965	9.925	22.328	31.600
3	1.638	2.353	3.182	4.541	5.841	10.214	12.924
4	1.533	2.132	2.776	3.747	4.604	7.173	8.610
5	1.476	2.015	2.571	3.365	4.032	5.894	6.869
6	1.440	1.943	2.447	3.143	3.707	5.208	5.959
7	1.415	1.895	2.365	2.998	3.499	4.785	5.408
8	1.397	1.860	2.306	2.896	3.355	4.501	5.041
9	1.383	1.833	2.262	2.821	3.250	4.297	4.781
10	1.372	1.812	2.228	2.764	3.169	4.144	4.587
11	1.363	1.796	2.201	2.718	3.106	4.025	4.437
12	1.356	1.782	2.179	2.681	3.055	3.930	4.318
13	1.350	1.771	2.160	2.650	3.012	3.852	4.221
14	1.345	1.761	2.145	2.624	2.977	3.787	4.140
15	1.341	1.753	2.131	2.602	2.947	3.733	4.073
16	1.337	1.746	2.120	2.583	2.921	3.686	4.015
17	1.333	1.740	2.110	2.567	2.898	3.646	3.965
18	1.330	1.734	2.101	2.552	2.878	3.610	3.922
19	1.328	1.729	2.093	2.539	2.861	3.579	3.883
20	1.325	1.725	2.086	2.528	2.845	3.552	3.850
21	1.323	1.721	2.080	2.518	2.831	3.527	3.819
22	1.321	1.717	2.074	2.508	2.819	3.505	3.792
23	1.319	1.714	2.069	2.500	2.807	3.485	3.768
24	1.318	1.711	2.064	2.492	2.797	3.467	3.745
25	1.316	1.708	2.060	2.485	2.787	3.450	3.725
26	1.315	1.706	2.056	2.479	2.779	3.435	3.707
27	1.314	1.703	2.052	2.473	2.771	3.421	3.689
28	1.313	1.701	2.048	2.467	2.763	3.408	3.674
29	1.311	1.699	2.045	2.462	2.756	3.396	3.660
30	1.310	1.697	2.042	2.457	2.750	3.385	3.646
32	1.309	1.694	2.037	2.449	2.738	3.365	3.622
34	1.307	1.691	2.032	2.441	2.728	3.348	3.601
36	1.306	1.688	2.028	2.434	2.719	3.333	3.582
38	1.304	1.686	2.024	2.429	2.712	3.319	3.566
40	1.303	1.684	2.021	2.423	2.704	3.307	3.551
50	1.299	1.676	2.009	2.403	2.678	3.261	3.496
75	1.293	1.665	1.992	2.377	2.643	3.202	3.425
100	1.290	1.660	1.984	2.364	2.626	3.174	3.390
∞	1.282	1.645	1.960	2.326	2.576	3.090	3.290

A4: Critical Values of the F Distribution at a 5 Percent Level of Significance



	Degrees of Freedom for the Numerator																			
	1	2	3	4	5	6	7	8	9	10	12	15	20	24	30	40	60	120	inf	
Degrees of Freedom for the Denominator	1	161.45	199.50	215.71	224.58	230.16	233.99	236.77	238.88	240.54	241.88	243.91	245.95	248.01	249.05	250.10	251.14	252.20	253.25	254.31
	2	18.51	19.00	19.16	19.25	19.30	19.33	19.35	19.37	19.38	19.40	19.41	19.43	19.45	19.45	19.46	19.47	19.48	19.49	19.50
	3	10.13	9.55	9.28	9.12	9.01	8.94	8.89	8.85	8.81	8.79	8.74	8.70	8.66	8.64	8.62	8.59	8.57	8.55	8.53
	4	7.71	6.94	6.59	6.39	6.26	6.16	6.09	6.04	6.00	5.96	5.91	5.86	5.80	5.77	5.75	5.72	5.69	5.66	5.63
	5	6.61	5.79	5.41	5.19	5.05	4.95	4.88	4.82	4.77	4.74	4.68	4.62	4.56	4.53	4.50	4.46	4.43	4.40	4.37
	6	5.99	5.14	4.76	4.53	4.39	4.28	4.21	4.15	4.10	4.06	4.00	3.94	3.87	3.84	3.81	3.77	3.74	3.70	3.67
	7	5.59	4.74	4.35	4.12	3.97	3.87	3.79	3.73	3.68	3.64	3.57	3.51	3.44	3.41	3.38	3.34	3.30	3.27	3.23
	8	5.32	4.46	4.07	3.84	3.69	3.58	3.50	3.44	3.39	3.35	3.28	3.22	3.15	3.12	3.08	3.04	3.01	2.97	2.93
	9	5.12	4.26	3.86	3.63	3.48	3.37	3.29	3.23	3.18	3.14	3.07	3.01	2.94	2.90	2.86	2.83	2.79	2.75	2.71
	10	4.96	4.10	3.71	3.48	3.33	3.22	3.14	3.07	3.02	2.98	2.91	2.85	2.77	2.74	2.70	2.66	2.62	2.58	2.54
	11	4.84	3.98	3.59	3.36	3.20	3.09	3.01	2.95	2.90	2.85	2.79	2.72	2.65	2.61	2.57	2.53	2.49	2.45	2.40
	12	4.75	3.89	3.49	3.26	3.11	3.00	2.91	2.85	2.80	2.75	2.69	2.62	2.54	2.51	2.47	2.43	2.38	2.34	2.30
	13	4.67	3.81	3.41	3.18	3.03	2.92	2.83	2.77	2.71	2.67	2.60	2.53	2.46	2.42	2.38	2.34	2.30	2.25	2.21
	14	4.60	3.74	3.34	3.11	2.96	2.85	2.76	2.70	2.65	2.60	2.53	2.46	2.39	2.35	2.31	2.27	2.22	2.18	2.13
	15	4.54	3.68	3.29	3.06	2.90	2.79	2.71	2.64	2.59	2.54	2.48	2.40	2.33	2.29	2.25	2.20	2.16	2.11	2.07
	16	4.49	3.63	3.24	3.01	2.85	2.74	2.66	2.59	2.54	2.49	2.42	2.35	2.28	2.24	2.19	2.15	2.11	2.06	2.01
	17	4.45	3.59	3.20	2.96	2.81	2.70	2.61	2.55	2.49	2.45	2.38	2.31	2.23	2.19	2.15	2.10	2.06	2.01	1.96
	18	4.41	3.55	3.16	2.93	2.77	2.66	2.58	2.51	2.46	2.41	2.34	2.27	2.19	2.15	2.11	2.06	2.02	1.97	1.92
	19	4.38	3.52	3.13	2.90	2.74	2.63	2.54	2.48	2.42	2.38	2.31	2.23	2.16	2.11	2.07	2.03	1.98	1.93	1.88
	20	4.35	3.49	3.10	2.87	2.71	2.60	2.51	2.45	2.39	2.35	2.28	2.20	2.12	2.08	2.04	1.99	1.95	1.90	1.84
	21	4.32	3.47	3.07	2.84	2.68	2.57	2.49	2.42	2.37	2.32	2.25	2.18	2.10	2.05	2.01	1.96	1.92	1.87	1.81
	22	4.30	3.44	3.05	2.82	2.66	2.55	2.46	2.40	2.34	2.30	2.23	2.15	2.07	2.03	1.98	1.94	1.89	1.84	1.78
	23	4.28	3.42	3.03	2.80	2.64	2.53	2.44	2.37	2.32	2.27	2.20	2.13	2.05	2.01	1.96	1.91	1.86	1.81	1.76
	24	4.26	3.40	3.01	2.78	2.62	2.51	2.42	2.36	2.30	2.25	2.18	2.11	2.03	1.98	1.94	1.89	1.84	1.79	1.73
	25	4.24	3.39	2.99	2.76	2.60	2.49	2.40	2.34	2.28	2.24	2.16	2.09	2.01	1.96	1.92	1.87	1.82	1.77	1.71
	26	4.23	3.37	2.98	2.74	2.59	2.47	2.39	2.32	2.27	2.22	2.15	2.07	1.99	1.95	1.90	1.85	1.80	1.75	1.69
	27	4.21	3.35	2.96	2.73	2.57	2.46	2.37	2.31	2.25	2.20	2.13	2.06	1.97	1.93	1.88	1.84	1.79	1.73	1.67
	28	4.20	3.34	2.95	2.71	2.56	2.45	2.36	2.29	2.24	2.19	2.12	2.04	1.96	1.91	1.87	1.82	1.77	1.71	1.65
	29	4.18	3.33	2.93	2.70	2.55	2.43	2.35	2.28	2.22	2.18	2.10	2.03	1.94	1.90	1.85	1.81	1.75	1.70	1.64
	30	4.17	3.32	2.92	2.69	2.53	2.42	2.33	2.27	2.21	2.16	2.09	2.01	1.93	1.89	1.84	1.79	1.74	1.68	1.62
	40	4.08	3.23	2.84	2.61	2.45	2.34	2.25	2.18	2.12	2.08	2.00	1.92	1.84	1.79	1.74	1.69	1.64	1.58	1.51
	60	4.00	3.15	2.76	2.53	2.37	2.25	2.17	2.10	2.04	1.99	1.92	1.84	1.75	1.70	1.65	1.59	1.53	1.47	1.39
	120	3.92	3.07	2.68	2.45	2.29	2.18	2.09	2.02	1.96	1.91	1.83	1.75	1.66	1.61	1.55	1.50	1.43	1.35	1.25
	inf	3.84	3.00	2.60	2.37	2.21	2.10	2.01	1.94	1.88	1.83	1.75	1.67	1.57	1.52	1.46	1.39	1.32	1.22	1.00

A5: Durbin-Watson d Statistic: Significance Points of dL and dU at 0.05 Level of Significance

k = numbers of parameters estimated in the model

n	k = 2		k = 3		k = 4		k = 5		k = 6		k = 7	
	dL	dU	dL	dU	dL	dU	dL	dU	dL	dU	dL	dU
6	0.610	1.400										
7	0.700	1.356	0.467	1.896								
8	0.763	1.332	0.559	1.777	0.367	2.287						
9	0.824	1.320	0.629	1.699	0.455	2.128	0.296	2.588				
10	0.879	1.320	0.697	1.641	0.525	2.016	0.376	2.414	0.243	2.822		
11	0.927	1.324	0.758	1.604	0.595	1.928	0.444	2.283	0.315	2.645	0.203	3.004
12	0.971	1.331	0.812	1.579	0.658	1.864	0.512	2.177	0.380	2.506	0.268	2.832
13	1.010	1.340	0.861	1.562	0.715	1.816	0.574	2.094	0.444	2.390	0.328	2.692
14	1.045	1.350	0.905	1.551	0.767	1.779	0.632	2.030	0.505	2.296	0.389	2.572
15	1.077	1.361	0.946	1.543	0.814	1.750	0.685	1.977	0.562	2.220	0.447	2.471
16	1.106	1.371	0.982	1.539	0.857	1.728	0.734	1.935	0.615	2.157	0.502	2.388
17	1.133	1.381	1.015	1.536	0.897	1.710	0.779	1.900	0.664	2.104	0.554	2.318
18	1.158	1.391	1.046	1.535	0.933	1.696	0.820	1.872	0.710	2.060	0.603	2.258
19	1.180	1.401	1.074	1.536	0.967	1.685	0.859	1.848	0.752	2.023	0.649	2.206
20	1.201	1.411	1.100	1.537	0.998	1.676	0.894	1.828	0.792	1.991	0.691	2.162
21	1.221	1.420	1.125	1.538	1.026	1.669	0.927	1.812	0.829	1.964	0.731	2.124
22	1.239	1.429	1.147	1.541	1.053	1.664	0.958	1.797	0.863	1.940	0.769	2.090
23	1.257	1.437	1.168	1.543	1.078	1.660	0.986	1.785	0.895	1.920	0.804	2.061
24	1.273	1.446	1.188	1.546	1.101	1.656	1.013	1.775	0.925	1.902	0.837	2.035
25	1.288	1.454	1.206	1.550	1.123	1.654	1.038	1.767	0.953	1.886	0.868	2.013
26	1.302	1.461	1.224	1.553	1.143	1.652	1.062	1.759	0.979	1.873	0.897	1.992
27	1.316	1.469	1.240	1.556	1.162	1.651	1.084	1.753	1.004	1.861	0.925	1.974
28	1.328	1.476	1.255	1.560	1.181	1.650	1.104	1.747	1.028	1.850	0.951	1.959
29	1.341	1.483	1.270	1.563	1.198	1.650	1.124	1.743	1.050	1.841	0.975	1.944
30	1.352	1.489	1.284	1.567	1.214	1.650	1.143	1.739	1.071	1.833	0.998	1.931
31	1.363	1.496	1.297	1.570	1.229	1.650	1.160	1.735	1.090	1.825	1.020	1.920
32	1.373	1.502	1.309	1.574	1.244	1.650	1.177	1.732	1.109	1.819	1.041	1.909
33	1.383	1.508	1.321	1.577	1.258	1.651	1.193	1.730	1.127	1.813	1.061	1.900
34	1.393	1.514	1.333	1.580	1.271	1.652	1.208	1.728	1.144	1.808	1.079	1.891
35	1.402	1.519	1.343	1.584	1.283	1.653	1.222	1.726	1.160	1.803	1.097	1.884
36	1.411	1.525	1.354	1.587	1.295	1.654	1.236	1.724	1.175	1.799	1.114	1.876
37	1.419	1.530	1.364	1.590	1.307	1.655	1.249	1.723	1.190	1.795	1.131	1.870
38	1.427	1.535	1.373	1.594	1.318	1.656	1.261	1.722	1.204	1.792	1.146	1.864
39	1.435	1.540	1.382	1.597	1.328	1.658	1.273	1.722	1.218	1.789	1.161	1.859
40	1.442	1.544	1.391	1.600	1.338	1.659	1.285	1.721	1.230	1.786	1.175	1.854
41	1.449	1.549	1.399	1.603	1.348	1.660	1.296	1.720	1.243	1.784	1.189	1.849
42	1.456	1.553	1.407	1.606	1.357	1.662	1.306	1.720	1.255	1.781	1.202	1.845
43	1.463	1.558	1.415	1.609	1.366	1.663	1.317	1.720	1.266	1.779	1.215	1.841
44	1.469	1.562	1.423	1.612	1.375	1.665	1.326	1.720	1.277	1.778	1.227	1.838
45	1.475	1.566	1.430	1.615	1.383	1.666	1.336	1.720	1.287	1.776	1.238	1.835
46	1.481	1.570	1.437	1.618	1.391	1.668	1.345	1.720	1.298	1.775	1.250	1.832
47	1.487	1.574	1.444	1.620	1.399	1.669	1.354	1.720	1.307	1.774	1.260	1.829
48	1.493	1.578	1.450	1.623	1.406	1.671	1.362	1.721	1.317	1.773	1.271	1.826
49	1.498	1.581	1.456	1.626	1.414	1.672	1.370	1.721	1.326	1.772	1.281	1.824
50	1.503	1.585	1.462	1.628	1.421	1.674	1.378	1.721	1.335	1.771	1.291	1.822
51	1.509	1.588	1.468	1.631	1.427	1.675	1.386	1.722	1.343	1.770	1.300	1.820
52	1.514	1.592	1.474	1.633	1.434	1.677	1.393	1.722	1.351	1.769	1.309	1.818
53	1.518	1.595	1.480	1.636	1.440	1.678	1.400	1.723	1.359	1.769	1.318	1.817
54	1.523	1.598	1.485	1.638	1.446	1.680	1.407	1.723	1.367	1.768	1.326	1.815
55	1.528	1.601	1.490	1.641	1.452	1.681	1.414	1.724	1.374	1.768	1.334	1.814
56	1.532	1.605	1.495	1.643	1.458	1.683	1.420	1.725	1.382	1.768	1.342	1.812
57	1.536	1.608	1.500	1.645	1.464	1.684	1.426	1.725	1.389	1.768	1.350	1.811
58	1.540	1.610	1.505	1.647	1.469	1.686	1.433	1.726	1.395	1.767	1.358	1.810
59	1.545	1.613	1.510	1.650	1.474	1.687	1.438	1.727	1.402	1.767	1.365	1.809
60	1.549	1.616	1.514	1.652	1.480	1.689	1.444	1.727	1.408	1.767	1.372	1.808
61	1.552	1.619	1.519	1.654	1.485	1.690	1.450	1.728	1.415	1.767	1.379	1.807
62	1.556	1.622	1.523	1.656	1.490	1.692	1.455	1.729	1.421	1.767	1.385	1.807
63	1.560	1.624	1.527	1.658	1.494	1.693	1.461	1.730	1.427	1.767	1.392	1.806
64	1.563	1.627	1.532	1.660	1.499	1.695	1.466	1.730	1.432	1.767	1.398	1.805
65	1.567	1.629	1.536	1.662	1.503	1.696	1.471	1.731	1.438	1.767	1.404	1.805
66	1.570	1.632	1.539	1.664	1.508	1.697	1.476	1.732	1.443	1.767	1.410	1.804
67	1.574	1.634	1.543	1.666	1.512	1.699	1.481	1.733	1.449	1.768	1.416	1.804
68	1.577	1.637	1.547	1.668	1.516	1.700	1.485	1.733	1.454	1.768	1.422	1.803
69	1.580	1.639	1.551	1.670	1.521	1.701	1.490	1.734	1.459	1.768	1.427	1.803
70	1.583	1.641	1.554	1.672	1.525	1.703	1.494	1.735	1.464	1.768	1.433	1.802
75	1.598	1.652	1.571	1.680	1.543	1.709	1.515	1.739	1.487	1.770	1.458	1.801
80	1.611	1.662	1.586	1.688	1.560	1.715	1.534	1.743	1.507	1.772	1.480	1.801
85	1.624	1.671	1.600	1.696	1.575	1.721	1.550	1.747	1.525	1.774	1.500	1.801
90	1.635	1.679	1.612	1.703	1.589	1.726	1.566	1.751	1.542	1.776	1.518	1.801
95	1.645	1.687	1.623	1.709	1.602	1.732	1.579	1.755	1.557	1.778	1.535	1.802
100	1.654	1.694	1.634	1.715	1.613	1.736	1.592	1.758	1.571	1.780	1.550	1.803
200	1.758	1.778	1.748	1.789	1.738	1.799	1.728	1.810	1.718	1.820	1.707	1.831

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